Cloud Computing and Virtual Networks

Cloud computing allows organizations to obtain computing resources over a network instead of purchasing and maintaining all physical infrastructure themselves. Cloud providers offer services for compute, storage, databases, networking, security, and analytics. Resources can often be created or removed on demand.

Virtual machines are one common form of cloud compute. A virtual machine provides an operating system environment running on physical infrastructure managed by the cloud provider. Virtual machines can be configured with different amounts of CPU, memory, and storage. Serverless computing is another approach in which developers deploy functions without directly managing servers.

Cloud storage is commonly divided into different categories. Object storage is designed for files and other large objects, while block storage provides disk-like storage that can be attached to compute resources. Managed databases reduce the amount of maintenance required from application teams.

Networking is an important part of cloud architecture. A virtual network provides an isolated networking environment for cloud resources. Subnets divide a virtual network into smaller network segments. Routing controls how traffic moves between networks, while security rules restrict which connections are allowed.

Identity and access management controls who can access cloud resources and what actions they can perform. Good cloud security generally follows least privilege, meaning users receive only the permissions they need. Encryption protects data while stored and while traveling across networks. Monitoring and logging help teams diagnose failures and understand system behavior.